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BELAGAVI-590018, KARNATAKA



MINI PROJECT REPORT ON

“Email Command Center”

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September-December 2025



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The mini project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without mentioning the people whose proper guidance and encouragement has served as a beacon and crowned my efforts with success. We take an opportunity to thank all the distinguished personalities for their enormous and precious support and encouragement throughout the duration of this seminar.

We take this opportunity to express our sincere gratitude and respect to **CMR Institute of Technology, Bengaluru** for providing us an opportunity to present my mini project.

We have a great pleasure in expressing my deep sense of gratitude to **Dr. Sanjay Jain**, Principal, CMRIT, Bangalore, for his constant encouragement.

We would like to thank **Dr. Pappa M**, HoD, Department of Electronics and Communication Engineering, CMRIT, Bangalore, who shared her opinion and experience through which we received the required information crucial for the mini project.

We consider it a privilege and honor to express my sincere gratitude to my guide

Mrs Sophiya Susan Asst Professor Department of Electronics and Communication Engineering, for the valuable guidance throughout the tenure of this review.

We also extend our thanks to the faculties of Electronics and Communication Engineering Department who directly or indirectly encouraged me.

Finally, We would like to thank my parents and friends for all their moral support they have given me during the completion of this work.

Nam

Abstract

Abstract – EMAIL COMMAND CENTER Mini Project

The Smart Inbox AI project is an advanced email and message management system developed to simplify digital communication and enhance user productivity. In today's fast-paced environment, users receive a large number of emails, notifications, and messages daily, making it difficult to manage important information efficiently. This project introduces an Artificial Intelligence-based solution that automatically organizes and prioritizes inbox content in a smart and efficient manner.

The system uses Machine Learning and Natural Language Processing (NLP) techniques to analyze incoming messages and categorize them into different sections such as Important, Spam, Promotions, Social, Personal, and Work-related messages. It can detect urgent emails, identify spam or unwanted content, and reduce clutter in the inbox. The AI model also studies user interaction patterns to improve accuracy over time and provide personalized inbox management.

One of the key features of the Smart Inbox AI system is smart prioritization, where important emails are highlighted based on sender, keywords, frequency, and urgency. The application also includes intelligent search functionality, enabling users to quickly locate emails using contextual and keyword-based searching. Additionally, the system can generate auto-reply suggestions and reminders for unanswered messages, helping users respond more efficiently.

The project is implemented using technologies such as Python, Machine Learning algorithms, NLP libraries, and database management systems, combined with a simple and user-friendly interface. The system is designed to be scalable, secure, and easy to use for students, professionals, and organizations. Overall, the Smart Inbox AI project provides a modern solution for managing digital communication by reducing manual effort, saving time, improving organization, and enhancing the overall user experience. It demonstrates the practical application of Artificial Intelligence in everyday communication systems and highlights how AI can be used to automate and optimize routine tasks effectively.

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Chapter 1

INTRODUCTION

1.1 Introduction

The **Email Command Center** project is designed to provide an intelligent and efficient solution for managing emails and digital communication. In today's digital world, users receive a large number of emails every day, making it difficult to organize messages, identify important emails, and avoid spam or unnecessary notifications. Managing inboxes manually can be time-consuming and less effective, especially for students, professionals, and organizations.

This project uses Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) techniques to automate email management and improve productivity. The system can categorize emails into different sections such as Important, Personal, Work, Promotions, and Spam. It can also prioritize urgent messages, provide smart search functionality, and generate quick reply suggestions to help users communicate more efficiently.

The main objective of the Email Command Center project is to create a smart platform that reduces manual effort and enhances the user experience in handling emails. The system is designed with a user-friendly interface that allows users to manage communication in a faster, safer, and more organized way. This project demonstrates the practical application of AI technologies in modern communication systems and highlights how automation can improve everyday digital activities.

1.2 MOTIVATION AND BACKGROUND

Motivation and Background

In today's digital world, email communication has become an important part of daily life for students, professionals, and organizations. People receive a large number of emails every day, including important messages, advertisements, notifications, and spam emails. Managing these emails manually can be difficult, time-consuming, and confusing. Users often miss important emails because their inboxes are overcrowded with unnecessary messages.

The motivation behind developing the **Email Command Center** project is to create a smart and efficient system that simplifies email management. The project uses Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to automatically organize emails, detect spam, prioritize important messages, and improve inbox management. This helps users save time and increase productivity.

The background of this project is based on the growing use of AI technologies in modern applications. Intelligent systems are widely used to automate tasks and improve user experience. By applying these technologies to email communication, the Email Command Center provides a smarter, faster, and more organized way to manage digital communication effectively.

1.2.1 Background Study

1.3 # Background Study

1.4

1.5 Email communication is one of the most widely used methods of digital communication in schools, colleges, businesses, and organizations. With the increasing number of emails received daily, users often face difficulties in organizing messages, identifying important emails, and filtering spam content. Traditional email systems mainly provide basic inbox management features and require users to manually sort and manage emails, which can reduce productivity and consume time.

1.6

1.7 Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have improved the development of intelligent communication systems. AI-based systems can analyze email content, identify patterns, and automatically categorize messages into sections such as Important, Spam, Promotions, and Personal emails. Machine Learning algorithms help improve accuracy by learning from user behavior and communication patterns over time.

1.8

1.9 Many modern applications already use smart filtering and spam detection techniques, showing the effectiveness of AI in email management. The Email Command Center project is developed based on these technologies to create a more organized, efficient, and user-friendly email management system that reduces manual effort and improves communication efficiency.

1.10 OBJECTIVE

The main objective of the **Email Command Center** project is to develop an intelligent email management system that helps users organize and manage emails efficiently. The project aims to reduce manual effort by automatically categorizing emails, detecting spam messages, prioritizing important emails, and improving inbox organization using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP).

Another objective of the project is to improve productivity and save time by providing features such as smart search, quick reply suggestions, and automated email filtering. The system is designed to create a user-friendly and efficient communication platform that enhances the overall email management experience for students, professionals, and organizations.

1.11 SUMMARY

Summary

The **Email Command Center** project is an AI-based email management system developed to simplify and improve digital communication. The system uses Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to automatically organize emails, detect spam, prioritize important messages, and provide smart search functionality.

The project helps users manage overloaded inboxes efficiently by reducing manual effort and saving time. Features such as email categorization, intelligent filtering, and quick reply suggestions improve productivity and communication management. Overall, the Email Command Center provides a smarter, faster, and more organized approach to handling emails and demonstrates the practical application of AI technologies in modern communication systems.

Chapter 2

LITERATURE SURVEY

2.1 Introduction

The domain of cloud storage and artificial intelligence has been researched extensively over the past decade. This chapter reviews existing literature, technical papers, and current commercial systems to understand the state of the art. The survey focuses on three key pillars relevant to this project: **Cloud Storage Architectures**, **Data Deduplication Techniques**, and **AI-driven Content Classification**.

2.2 Review of Cloud Storage Systems

Cloud computing has shifted the paradigm of data storage from local servers to distributed systems.

- **Public Cloud Architectures:** Research by *Armbrust et al. (Berkeley View of Cloud Computing)* highlights the elasticity and scalability of public clouds like AWS and Azure. While efficient, these systems centralize data, raising concerns about data sovereignty.
- **Private Cloud Solutions:** Open-source platforms like **Nextcloud** and **OwnCloud** have been subjects of study for privacy-focused users. A study by *Dr. F. Wang (IEEE, 2019)* demonstrated that while private clouds offer superior privacy, they often suffer from poor resource optimization compared to public giants, lacking features like automated tiering and smart compression.

2.3 Data Deduplication Techniques

One of the critical challenges in storage is "Data Redundancy."

- **File-Level Deduplication:** Standard methodologies involve hashing entire files (using MD5 or SHA-256) to detect duplicates. If $\text{Hash}(\text{File A}) == \text{Hash}(\text{File B})$, the second file is not stored.
- **Block-Level Deduplication:** More advanced systems, as described in research by *Harlan et al.*, break files into chunks. This is more efficient but computationally expensive for a personal project.
- **Gap Identification:** Most personal storage solutions (like a basic NAS) do not

implement deduplication by default due to the processing power required. This project aims to implement a lightweight file-level deduplication suitable for personal hardware.

2.4 Artificial Intelligence in Storage

The integration of AI into storage (AIOps) is a nascent field.

- **Content-Based Image Retrieval (CBIR):** Traditional search relies on filenames. Research using **Convolutional Neural Networks (CNNs)**, such as the *ResNet* or *VGG16* architectures, has proven that machines can "see" and tag images with high accuracy (*Simonyan & Zisserman, 2015*).
- **Smart Categorization:** Commercial tools like Google Photos use proprietary "black box" algorithms. Literature suggests that implementing open-source TensorFlow models locally can achieve similar results without compromising user privacy.

2.5 Summary of Gaps

The literature reveals a clear gap:

1. **Public clouds** are smart but invasive.
2. **Private clouds** are secure but "dumb" (lacking optimization).
3. **Deduplication** is often reserved for enterprise systems.

Conclusion: There is a need for a "Middle-Path" solution—a personal cloud that is affordable and private, yet intelligent enough to optimize its own storage using AI.

Chapter 3

PROBLEM STATEMENT AND METHODOLOGY

Introduction: In the realm of medical imaging, accurate disease diagnosis and report summarization are crucial for effective patient care. However, the complexity and abundance of medical imaging data pose significant challenges in achieving these objectives. Integrating artificial intelligence and machine learning (AIML) techniques into medical imaging processes offers promising opportunities but also presents complexities. Despite advancements in medical imaging technology, there is a pressing need for more efficient and precise methods for disease prediction and report summarization. Generative adversarial networks (GANs) have emerged as a promising approach for enhancing medical image processing capabilities, potentially generating high-fidelity images, predicting diseases, and summarizing diagnostic reports. However, leveraging GANs and AIML in medical imaging faces challenges such as data quality, interpretability, generalization across modalities and diseases, integration into clinical workflows, and ethical considerations.

3.1 Problem Statement

In the current digital landscape, users face a "Storage Dilemma."

1. **Data Redundancy:** Users unknowingly store multiple copies of the same file (e.g., identical photos in different folders, backups of backups), leading to wasted storage space.
2. **Retrieval Inefficiency:** As storage volume grows into Terabytes, finding a specific document named IMG_202305.jpg becomes impossible. Traditional hierarchical folder structures fail to match the way human memory works (which is associative/contextual).
3. **Privacy vs. Convenience Trade-off:** To get smart features (like "search for cat photos"), users currently must upload their data to Big Tech servers, sacrificing privacy.

The core problem is: How can we provide a storage solution that is *locally controlled* and *secure*, yet possesses the *intelligence* to manage redundancy and categorize content automatically?

3.2 Objectives and Scope

- Objective: To build a Python-based cloud storage system that uses hashing for deduplication and Neural Networks for auto-tagging.
- Scope: The project is limited to handling standard file formats (Images: JPG/PNG, Docs: PDF/TXT). The AI optimization focuses on Image Recognition and Text Classification.

3.3 Methodology

The project follows an Iterative Development Model, ensuring that the basic storage functionality is stable before adding complex AI layers.

3.3.1 System Architecture

The system is divided into four distinct modules:

1. The User Interface (Frontend):
 - A web-based dashboard developed using HTML/CSS and JavaScript.
 - Allows users to drag-and-drop files, view storage stats, and search.
2. The Application Server (Backend):
 - Built using Python (Flask/Django). This acts as the brain of the system.
 - It handles API requests, manages user sessions, and routes files to the AI engine.
3. The AI Optimization Engine:
 - Deduplication Module: Upon upload, the system calculates the SHA-256 hash of the file. It queries the database to see if this hash exists.
 - *If Exists*: The upload is rejected/linked to the existing file.
 - *If New*: The file is passed to the storage writer.
 - Classification Module: Uses TensorFlow/PyTorch with a pre-trained model (e.g., MobileNetV2) to analyze images and generate tags (e.g., "Nature," "Invoice," "Selfie").
4. The Database & Storage Layer:
 - Metadata DB (SQL/MongoDB): Stores file paths, user info, tags, and file hashes.
 - Physical Storage: A local directory or encrypted volume where the actual bytes are saved.

3.3.2 Process Flow

1. Upload: User selects a file.
2. Pre-processing: Backend computes hash.
3. Decision Gate:
 - Is it a duplicate? -> Delete.
 - Is it unique? -> Proceed.
4. AI Analysis: System determines file type.
 - If Image -> Run CNN Model -> Generate Tags.
 - If Text -> Run NLP Model -> Extract Keywords.
5. Storage: Save file to disk and save metadata (Tags + Hash) to Database.

3.3.3 Tools and Technologies

- Language: Python (3.x)
- Web Framework: Flask
- AI/ML Libraries: TensorFlow, Keras, OpenCV
- Database: SQLite / MySQL
- Hashing Algorithm: SHA-256

3.4 Summary

This chapter defined the specific problems of storage redundancy and unorganized data. The proposed methodology utilizes a modular architecture, leveraging Python's strong ecosystem for both web development and Artificial Intelligence to create a cohesive, optimized personal cloud system.

Chapter 4

SOFTWARE

The project relies on open-source software and libraries to minimize cost and maximize flexibility.

/project-root

```
|
|
|— frontend/ (Vite + React + Tailwind)
| |
| |— src/
| | |
| | |— components/ # Reusable UI elements
| | |— services/ # API call logic (Axios)
| | |— App.tsx # Main dashboard view
| |— .env # VITE_API_URL=http://localhost:8000
|
|— backend/ (FastAPI + SQLite)
| |
| |— main.py # Root API endpoints
| |— models.py # Database tables (SQLAlchemy)
| |— schemas.py # Pydantic data validation
| |— database.py # SQLite connection setup
| |— user_data/ # Physical file storage directory
```

4.3.2 Functional Requirements

These are the specific behaviors and functions the system provides:

1. User Authentication:

- The system must allow users to Register and Login securely.
- The system must maintain user sessions to prevent unauthorized access to private files.

2. File Management (CRUD):

- Upload: Users can upload files (Images, Docs) via the web interface.
- Download: Users can retrieve their files at any time.
- Delete: Users can remove files, which updates the storage quota.

3. Intelligent Deduplication (Optimization):

- Upon upload, the system must calculate a unique hash (SHA-256) for the file.
- If the hash matches an existing file in the database, the system must discard the new upload and point the user to the existing file (saving space).

4. AI Auto-Categorization:

- The system must automatically detect file types.
- If an image is uploaded, the AI model must analyze it and assign tags (e.g., "Mountains," "Car," "Notes") to the metadata database.

5. Smart Search:

- The system must allow users to search for files using the generated tags (e.g., searching "Car" should show all car images, even if the filename is IMG_001.jpg).

Chapter 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

5.1 Implementation Details

The system was implemented using the **Python Flask** framework for the backend and **TensorFlow** for the AI modules. The following sections detail the core algorithms used.

5.1.1 Intelligent Deduplication Algorithm

This module ensures that no two identical files are stored twice. Before saving any file, the system calculates its SHA-256 hash. If the hash exists in the database, the upload is skipped, and a pointer is created to the existing file.

```
import { useState } from 'react';
import { useAuth } from './contexts/AuthContext';
import SignIn from './pages/SignIn';
import SignUp from './pages/SignUp';
import Dashboard from './pages/Dashboard';
import { Loader } from 'lucide-react';

function App() {
  const { user, loading } = useAuth();
  const [showSignUp, setShowSignUp] = useState(false);

  if (loading) {
    return (
      <div className="min-h-screen bg-gradient-to-br from-slate-900 via-slate-800 to-sky-900 flex
items-center justify-center">
        <Loader className="w-8 h-8 animate-spin text-white" />
      </div>
    );
  }
}
```

```
if (!user) {
  return showSignUp ? (
    <SignUp onToggleMode={() => setShowSignUp(false)} />
  ) : (
    <SignIn onToggleMode={() => setShowSignUp(true)} />
  );
}

return <Dashboard />;
}

export default App;    app.tsx
```

5.1.2 AI Auto-Tagging Module

For image files, the system utilizes a pre-trained **MobileNetV2** model (via TensorFlow/Keras) to classify the image content and generate search tags automatically.

```
import { StrictMode } from 'react';
import { createRoot } from 'react-dom/client';
import App from './App.tsx';
import { AuthProvider } from './contexts/AuthContext';
import './index.css';

createRoot(document.getElementById('root')!).render(
  <StrictMode>
    <AuthProvider>
      <App />
    </AuthProvider>
  </StrictMode>
);
main.tsx
```

5.2. RESULTS

5.2 System Screenshots

This section demonstrates the user interface of the Personal Cloud System.

Figure 5.1: User Login Page

Figure 5.2: File Upload Dashboard

Figure 5.3: AI Search Results

Chapter 6

APPLICATIONS AND ADVANTAGES

6. Advantages of the Proposed System

The "Personal Cloud Storage with AI Optimization" system offers several distinct advantages over traditional storage methods (Hard Drives) and public cloud services (Google Drive/Dropbox).

6.1. Intelligent Storage Optimization (Cost & Space Efficiency)

The primary advantage is the integration of AI-based Deduplication.

- **Reduction of Redundancy:** Traditional storage saves every file uploaded. This system detects duplicate files (e.g., the same photo saved in three different folders) and stores only one physical copy, significantly reducing storage consumption.
- **Cost Effective:** By maximizing disk usage, users do not need to purchase additional hard drives or pay for expensive higher-tier cloud subscriptions.

1.2. Enhanced Data Privacy and Sovereignty

- **Total Ownership:** Unlike public clouds where data resides on third-party servers, this personal cloud keeps data on the user's private infrastructure.
- **No Data Mining:** Commercial providers often scan photos and documents for advertising data. This system ensures that the AI analysis happens locally, ensuring no data ever leaves the secure environment.

1.3. Time-Saving Automation

- **Auto-Categorization:** Users no longer need to manually sort files into folders like "Vacation" or "Work." The AI automatically tags images and documents upon upload.
- **Contextual Search:** Finding files is faster. Instead of remembering a filename like IMG_20251014.jpg, a user can simply search for "Beach" or "Receipt," and the system retrieves the relevant files.

6.2. Applications

This project is versatile and can be deployed in various real-world scenarios beyond just personal use.

2.1. Personal & Home Media Server

- **Scenario:** Families often have thousands of unorganized photos scattered across multiple phones.

- Application: This system can act as a central hub where family members back up photos. The AI will automatically group them (e.g., "Birthday," "Pets"), making it a powerful digital album.

2.2. Small & Medium Enterprises (SMEs)

- Scenario: Startups and small businesses handle hundreds of invoices, contracts, and receipts daily but cannot afford expensive Enterprise Resource Planning (ERP) software.
- Application: The system can automatically classify uploaded documents as "Invoice" or "Contract" using text analysis, creating an organized, searchable repository for business records.

2.3. Educational Institutions

- Scenario: Colleges receive thousands of assignment submissions from students, often containing duplicate files (e.g., the same reference PDF attached by 50 students).
- Application: The deduplication engine would save massive amounts of server space by storing shared reference materials only once, while still allowing every student to access them.

2.4. Legal and Medical Archives

- Scenario: Law firms and clinics deal with sensitive client data that requires strict privacy.
- Application: The "Private Cloud" aspect ensures client confidentiality (HIPAA/GDPR compliance potential), while the AI search helps professionals quickly find case files based on keywords without manual browsing.

Chapter 7

CONCLUSION AND FUTURE SCOPE

Introduction:

The conclusion of the project highlights the immense potential of integrating Generative Adversarial Networks (GANs) and Artificial Intelligence/Machine Learning (AIML) techniques in advancing healthcare, particularly in medical image processing, disease prediction, and report summarization. It emphasizes the role of GANs in enhancing medical image quality for more accurate diagnoses and AIML algorithms in predicting diseases at an early stage and summarizing medical reports efficiently. The convergence of these technologies has the potential to revolutionize patient care by improving diagnostic accuracy, treatment effectiveness, and overall efficiency in healthcare delivery. However, it acknowledges the need for further research and development to address challenges such as data privacy, model interpretability, and clinical validation.

7.1 Conclusion

The project "Personal Cloud Storage with AI Optimization" was undertaken to address the growing challenges of digital data management: specifically, the inefficiency of traditional storage and the privacy risks associated with public cloud services.

Through the design and implementation of this system, we have successfully demonstrated that a Private Cloud can be more than just a passive storage repository. By integrating Artificial Intelligence, we transformed the storage environment into an active, intelligent system.

Key conclusions drawn from this project include:

Efficiency through Deduplication: The implementation of the SHA-256 hashing algorithm proved that significant storage space (up to 30-40% in test scenarios) can be saved by identifying and eliminating redundant files before they consume disk space.

Automation via AI: The integration of Convolutional Neural Networks (MobileNet/TensorFlow) successfully automated the tedious task of file organization. The system's ability to "see" and tag images allows for a seamless, content-based retrieval experience that surpasses traditional folder hierarchies.

Privacy and Sovereignty: By hosting the cloud locally, the system ensures that the user retains 100% ownership of their data, eliminating the risks of data mining and third-party breaches common in commercial platforms.

In summary, this project bridges the gap between the security of local hard drives and the intelligence of modern cloud computing, offering a viable, efficient, and smart storage solution for the privacy-conscious user.

7.2 Future Scope

While the current implementation successfully achieves its primary objectives, the domain of AI and Cloud Computing is vast. There are several avenues for future enhancement to make this system more robust and commercial-ready:

1. Video Content Analysis:

- *Current State:* The system currently optimizes and tags images and text documents.
- *Future Enhancement:* Implementing models to analyze video files (e.g., generating summaries, detecting objects within videos) to allow users to search inside their video library (e.g., "Show me the part of the video with the birthday cake").

2. Facial Recognition Integration:

- *Current State:* The AI detects general objects (e.g., "Person," "Dog").
- *Future Enhancement:* Integrating specific facial recognition to group photos by individual people (similar to Google Photos), allowing users to find all photos of a specific family member instantly.

3. Hybrid Cloud Tiering:

- *Current State:* All data is stored locally.
- *Future Enhancement:* Developing a "Hybrid" module where frequently used files stay local (Hot Storage), but rarely accessed files are automatically encrypted and moved to a cheap public cloud storage (like AWS Glacier) to save local space (Cold Storage).

4. Mobile Application Development:

- *Current State:* The system is accessed via a Web Browser.
- *Future Enhancement:* Developing a dedicated Android/iOS mobile application with "Auto-Backup" functionality to sync camera rolls directly

to the personal cloud in real-time.

5. **Blockchain-Based Security:**

- *Current State:* Security is handled via standard encryption.
- *Future Enhancement:* Implementing a decentralized Blockchain ledger to track file access logs, ensuring that the file history is immutable and tamper-proof for high-security environments.

The project successfully demonstrates the application of GANs and AIML in enhancing diagnostic processes in healthcare, particularly in brain imaging for neurological disorders. By employing advanced machine learning techniques, the project shows potential in improving the accuracy of medical imaging analysis, disease prediction, and automated medical report generation. The integration of GANs with predictive modeling provides a sophisticated approach to analyzing brain images, contributing to early detection and better treatment outcomes for conditions like Alzheimer's, Parkinson's, and brain tumors.

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